Hints: Functional and differential equations – 1. Functional equations: the toolkit

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. First, $f \leqslant 1$: if $f(x) > 1$, the choice $y = \frac{x}{f(x)-1}$ makes $yf(x) = x + y$, and the equation gives $f(x)f(x+y) = f(x+y)$, i.e. $f(x) = 1$. Hence $f(x+y) = f(x)f(yf(x)) \leqslant f(x)$: f is non-increasing. If $f(x_0) = 1$ somewhere, then $f(x_0 + y) = f(y)$, so f is periodic and monotone, hence $f \equiv 1$.

Hint 2. Otherwise $f < 1$ everywhere, so $f(x+y) < f(x)$: f is strictly decreasing, in particular injective. Now split $x + y$ differently: $x + y = yf(x) + \big(x + y(1 - f(x))\big)$, and apply the equation to this splitting.

Hint 3. You get $f(x)f(yf(x)) = f(x+y) = f(yf(x))\, f\big((x + y(1 - f(x)))f(yf(x))\big)$, so by injectivity $x = \big(x + y(1 - f(x))\big)f(yf(x))$. Put $x = 1$ and $z = yf(1)$: $f(z) = \frac{1}{1+az}$ with $a = \frac{1-f(1)}{f(1)} \geqslant 0$. Answer: $f(x) = \frac{1}{1+ax}$, $a \geqslant 0$ ($a = 0$ is $f \equiv 1$); these work.

Problem 2

Hint 1. There is some z with $f(f(z)) > 0$. Otherwise $y \leqslant 0$ gives $f(x+y) \geqslant f(x)$, so f is non-increasing; then $f(x) \leqslant 0$ would give $f(f(x)) \geqslant f(0) > 0$, so $f > 0$ everywhere and $f(f(x)) > 0$ after all.

Hint 2. With such a z, $f(x) \geqslant f(z) + (x - z)f(f(z)) \to \infty$ as $x \to \infty$, and applying this to $f(x)$ in place of x gives $f(f(x)) \to \infty$. So the slopes in $f(x+y) \geqslant f(x) + yf(f(x))$ become arbitrarily large: $\frac{f(x)}{x} \to \infty$, and $f(x) > x + 1$ for large x.

Hint 3. For such an x take $y = f(x) - x > 0$: $f(f(x)) \geqslant f(x) + (f(x) - x)f(f(x))$, i.e. $f(f(x))\,(x + 1 - f(x)) \geqslant f(x)$. The left-hand side is negative ($f(f(x)) > 0$, $x + 1 - f(x) < 0$) and the right-hand side positive: a contradiction.

Problem 3

Hint 1. First f is injective: if $f(a) = f(b)$, $a < b$, then all terms f^k, $k \geqslant 1$, coincide at a and b, and $S = \mathrm{id} + f + \cdots + f^m$ has $S(a) < S(b)$. A continuous injective function is strictly monotone; if f were increasing, S would be increasing. So f is strictly decreasing, and the even iterates f^{2j} are increasing.

Hint 2. Group the terms of S in pairs: $f^{2j} + f^{2j+1} = g \circ f^{2j}$ with $g = \mathrm{id} + f$ non-decreasing, so each pair is non-decreasing. If m is even, the extra term f^m is strictly increasing, and S is strictly increasing: impossible.

Hint 3. So m is odd and $S = \sum_j g \circ f^{2j}$ is both non-increasing and a sum of non-decreasing functions; hence every summand is constant, in particular $g = \mathrm{id} + f$. Answer: $f(x) = c - x$ (with any odd m).

Problem 4

Hint 1. At $x = 0$: $f(0)^n \leqslant an^k$ for all n, so $f(0) = 1$. The answer is $f \equiv 1$: exponential growth beats an^k.

Hint 2. Suppose $f(x_0) > 1$, $x_0 \neq 0$. By continuity $f \geqslant q > 1$ on an interval $[x_0(1-\delta), x_0(1+\delta)]$. Take $x = \frac{x_0}{N}$ and $n = 2N$: the factors $f(jx)$ with $N(1-\delta) \leqslant j \leqslant N(1+\delta)$ are all $\geqslant q$, and there are about $2\delta N$ of them; all other factors are $\geqslant 1$.

Hint 3. So $q^{2\delta N - 1} \leqslant a(2N)^k$ for all N, which is false for large N. Answer: $f \equiv 1$.

Problem 5

Hint 1. On $\mathbb{Q}$: $f(0) = 0$, $f(nx) = nf(x)$ by induction, $f(-x) = -f(x)$, and then $qf(\frac{p}{q}) = f(p) = pf(1)$: $f(x) = cx$.

Hint 2. A monotone (say non-decreasing) solution on $\mathbb{R}$ is cx on $\mathbb{Q}$; for real x squeeze it between rationals $r < x < s$: $cr \leqslant f(x) \leqslant cs$, and let $r, s \to x$.

Hint 3. A Hamel basis B of $\mathbb{R}$ over $\mathbb{Q}$ exists by Zorn's lemma; prescribe f on B arbitrarily (say 1 at one basis element, 0 at the others) and extend $\mathbb{Q}$-linearly. Such an f is additive but not linear, its graph is dense, and it is not measurable. The school argument breaks at the passage from $\mathbb{Q}$ to $\mathbb{R}$: it needs a limit, i.e. some regularity.